

Short-term and long-term effects of AAV2-sVEGFR-2-Fc and AAV2-EGFP in the mouse eye following intravitreal injection

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Abstract

Background: Pathological angiogenesis in the eye is one of the hallmarks of vision-threatening diseases such as age-related macular degeneration. Vascular endothelial growth factors (VEGF) and their receptors are the key mediators of angiogenesis, making them potential targets in neovascular ocular diseases. Thus, anti-VEGF agents have revolutionized the treatment of these diseases. However, frequent intravitreal (IVT) injections of these agents place a burden on the patients and healthcare. Alternative therapies are urgently needed, and gene therapy offers a promising strategy. In the previous study, we showed that AAV2-mediated gene delivery of the truncated form of soluble VEGF receptor 2 (AAV2-sVEGFR-2-Fc) inhibits ocular neovascularization in a laser induced choroidal neovascularization mouse model. In the present study, we evaluate short-term and long-term gene expression, tropism, biodistribution, and safety of two different doses of AAV2-sVEGFR-2-Fc and AAV2-EGFP following IVT injection into the mouse eye.

Methods: Intraocular pressure (IOP) measurement, optical coherence tomography (OCT), and fundus imaging were performed at different timepoints after the IVT injection of AAV2-sVEGFR2 and -EGFP and animals were sacrificed. Viral vector biodistribution was determined by qPCR, and EGFP expression was assessed from the anti-GFP stained eye samples. OCT- and fundus images, and histological eye samples were used to evaluate the morphology and inflammation in the eye.

Results: Viral vector copies were predominantly found in the eyes and optic nerves and at low levels in off-target tissues such as liver. The EGFP transduction was mainly observed in ganglion cell layer 2 weeks after the AAV2-EGFP IVT injection. However, more widespread EGFP expression in the inner- and outer nuclear layer was observed 12 months after the AAV2-EGFP IVT Injection. Cell infiltration into the vitreous was observed following the IVT injection of both vectors with increased infiltration at higher doses, particularly at the 2-week timepoint. Less cell infiltration was prominent at later timepoints. All findings were dose dependent.

Conclusion: This study demonstrates that IVT injection of AAV2-mediated gene delivery can sustain long-term transgene expression in the eye. However, higher doses may induce inflammatory responses. Therefore, determining the optimal dose for AAV2-mediated ocular gene therapy is crucial to ensure both safety and therapeutic efficacy.